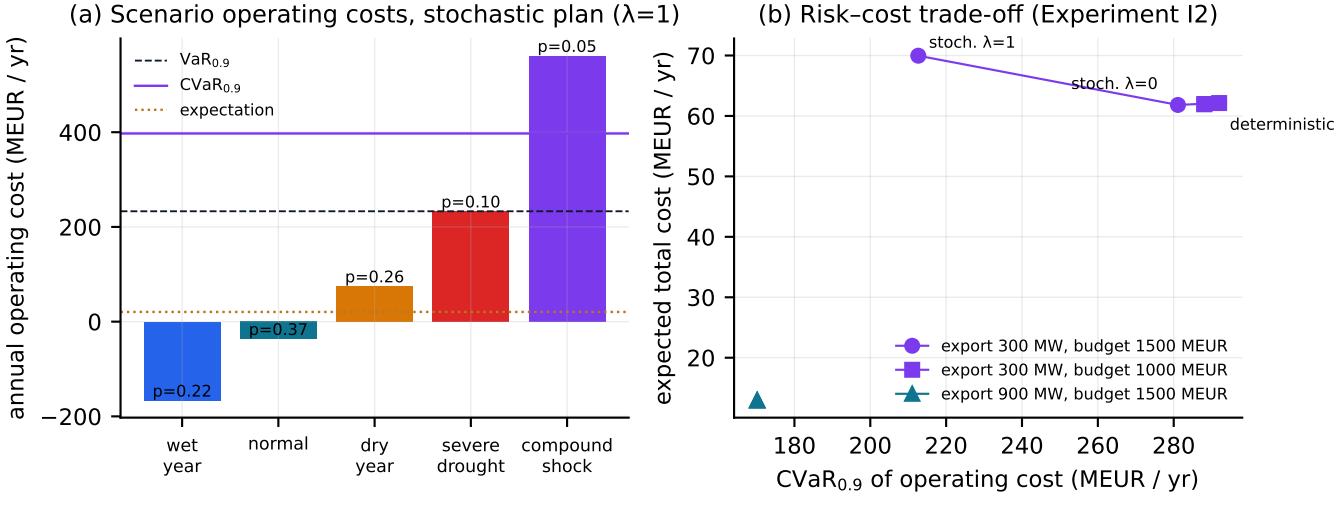
(a) Scenario operating costs, stochastic plan (λ=1)
VaR0.9
CVaR0.9
expectation
p=0.05
p=0.10
p=0.26
p=0.37
p=0.22
annual operating cost (MEUR / yr)
400
200
0
−200
wet year
normal
dry year
severe drought
compound shock
(b) Risk–cost trade-off (Experiment I2)
stoch. λ=1
stoch. λ=0
deterministic
export 300 MW, budget 1500 MEUR
export 300 MW, budget 1000 MEUR
export 900 MW, budget 1500 MEUR
expected total cost (MEUR / yr)
70
60
50
40
30
20
180
200
220
240
260
280
CVaR0.9 of operating cost (MEUR / yr)